

SYED MUJAHID

AEROSPACE DESIGN PORTFOLIO

UAV and eVTOL design | Aircraft CAD modeling
Engineering drawings | Structures and prototyping

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ABOUT ME

Concept to buildable aircraft hardware

I am an aeronautical engineer (B.E., Global Academy of Technology, 2024) specializing in aerospace conceptual design, CAD modeling, mechanical architecture, and prototyping. I focus on practical, manufacturable solutions, taking ideas from concept through detailed CAD and structural checks (FEA) into 3D-printable prototypes.

Beyond core design work, I coordinate with vendors and cross-functional teams and maintain technical documentation for advanced aircraft development programs, so designs stay clear and communicable across every phase of a project.

My work spans complete aircraft conceptual design, mission analysis, and system-level mechanical development, from a 40 kg modular VTOL UAV to a 19-passenger amphibious aircraft and full-scale eVTOL air ambulances.

I follow structured methods inspired by Mohammad H. Sadraey's *Aircraft Design: A Systems Engineering Approach*, balancing aerodynamics, structural integrity, and manufacturing practicality.

Currently: UAV prototype development at Dragale Aviation (Sep 2026 - present). That work is confidential and is not shown here; I am happy to discuss it in conversation.

TOOLKIT

CAD
SolidWorks, Autodesk Fusion 360, CATIA

Aero and visualization
OpenVSP, Blender

Analysis
Basic FEA, CFD, weight and CG estimation

Design methods
DfAM, DfA, mission analysis

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01 | DRAGALE AVIATION | 2026

Medical Transport VTOL UAV

Concept and full-scale CAD of a fixed-wing VTOL UAV for time-critical medical cargo.

46 kg
MTOW

16 kg
design payload

VTOL
fixed wing + lift rotors

Medical
organ-box payload

ROLE Design Engineer, Dragale Aviation (Apr - Jun 2026)
TOOLS Fusion 360, CFD

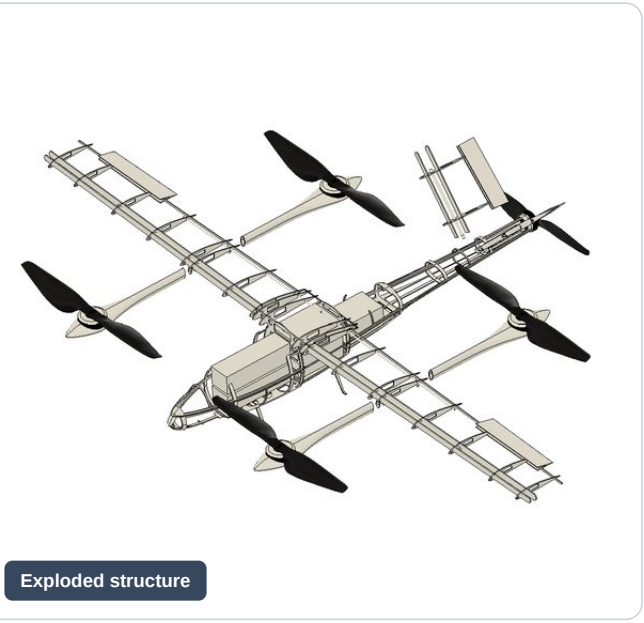
WHAT I DID

- Developed the concept and full-scale Fusion 360 CAD model of a 46 kg MTOW medical transport VTOL UAV with a 16 kg design payload.
- Sized the wing, fuselage, tail, propulsion system, battery, and landing gear.
- Integrated motors, propellers, batteries, and the payload bay into one airframe.
- Produced weight estimates, configuration trade studies, and CFD during concept development.





Full-scale CAD model



Exploded structure

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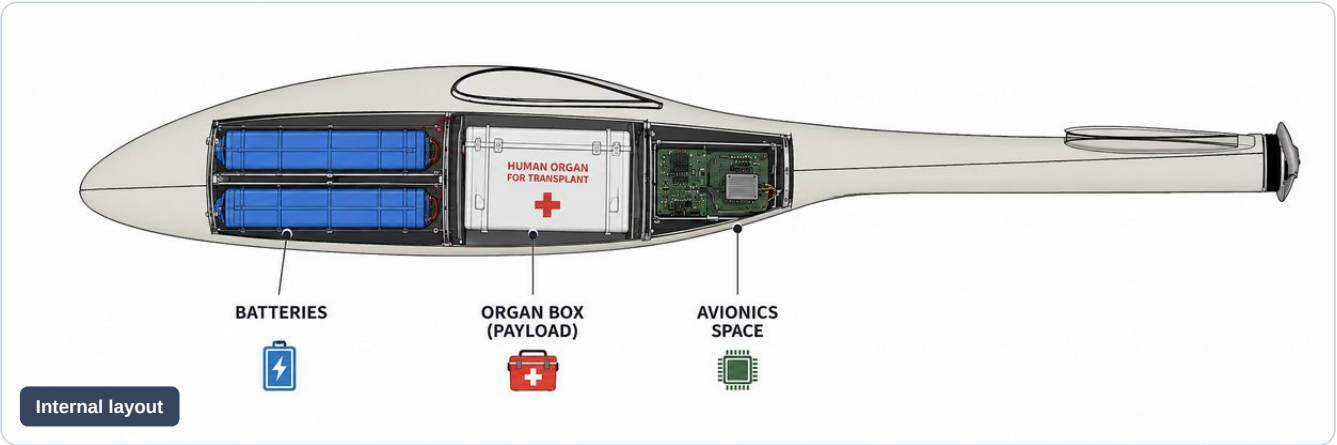
Internal layout and CFD

Payload packaging and aerodynamic analysis from the concept stage.

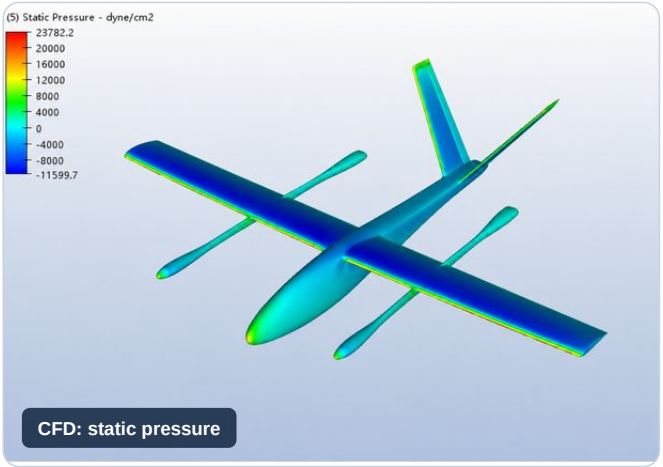
TOOLS Fusion 360, CFD

HIGHLIGHTS

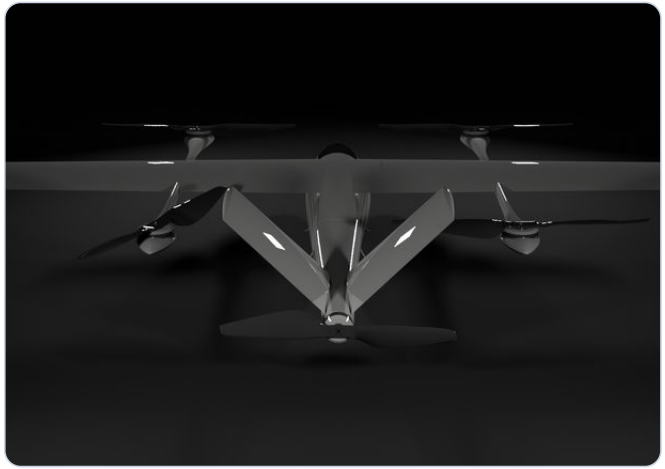
- **Internal layout:** battery, payload box, and avionics bays arranged along the fuselage.
- **CFD:** static pressure distribution over the airframe, used to support configuration trade studies.
- **Integration:** motors, propellers, batteries, and the payload bay packaged into one airframe.

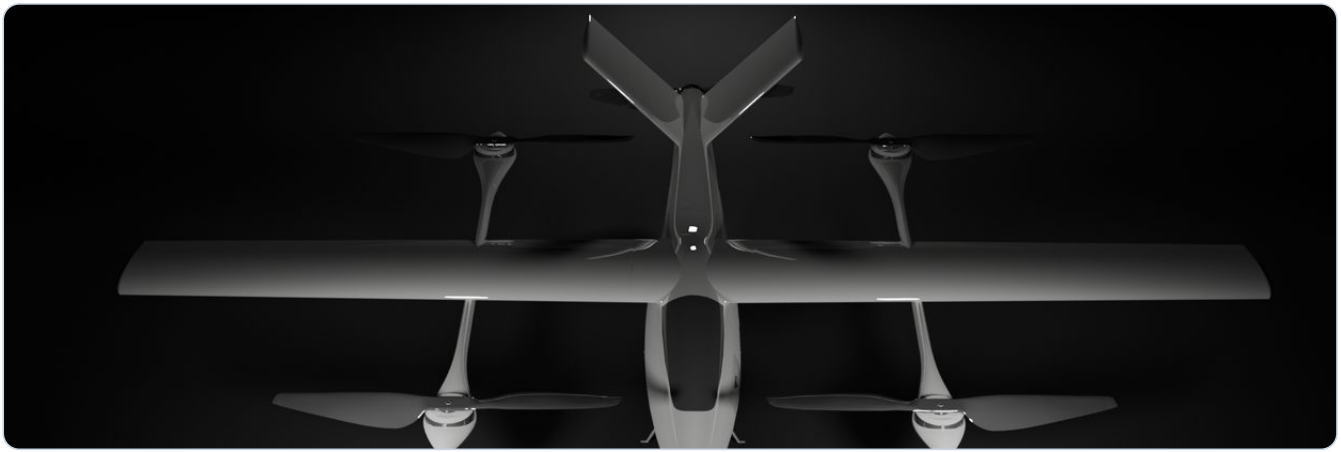


Internal layout



CFD: static pressure





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02 | DRAGALE AVIATION | 2026

Full-Scale Air Ambulance eVTOL

A three-person, 12-rotor eVTOL air ambulance modeled as a full-scale aircraft with system assemblies.

3
occupants

12
rotors

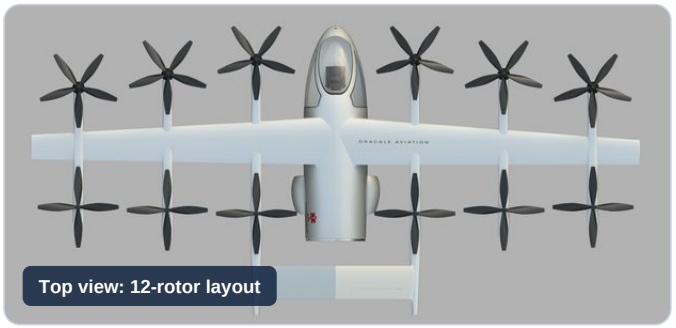
6
front tilt rotors

Rear
stretcher-loading door

ROLE Design Engineer, Dragale Aviation (Jul - Aug 2026)
TOOLS SolidWorks

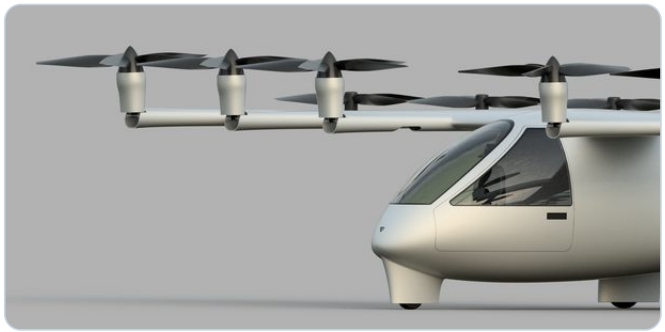
WHAT I DID

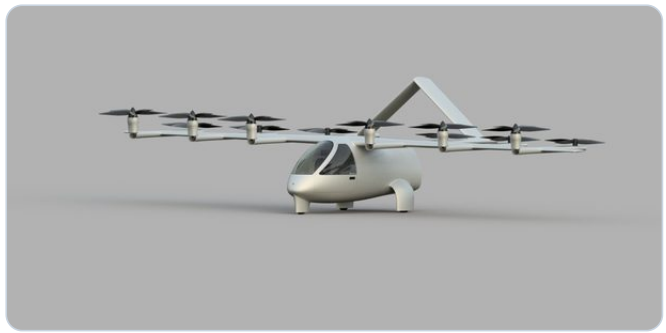
- Built a full-scale SolidWorks CAD model and system assemblies.
- Modeled the 12-rotor layout, front six-tilt-rotor mechanism, inverted V-tail boom mount, cabin layout, and rear stretcher-loading door.
- Compared configurations and subsystem layouts to assess feasibility and communicate the concept.


Top view: 12-rotor layout











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Modular 3D-Printable VTOL UAV

Fixed-wing VTOL UAV designed for additive manufacturing, from concept CAD to buildable parts.

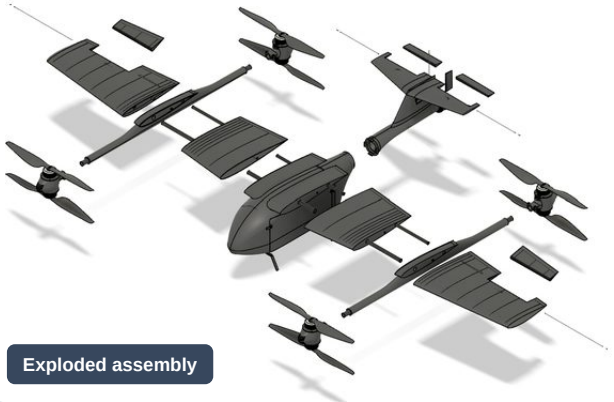
40 kg MTOW	**7 kg** design payload
8+1 coaxial propulsion	**Fusion 360** CAD

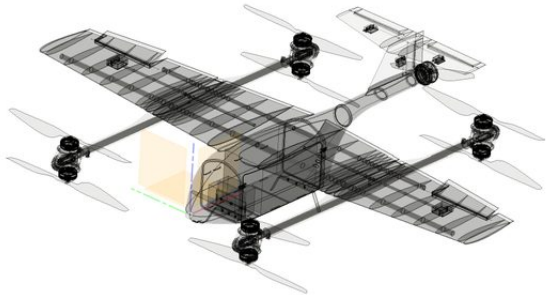
ROLE Design Engineer, Zayn Aerospace (Jan - Mar 2026, remote)
TOOLS Fusion 360, OpenVSP

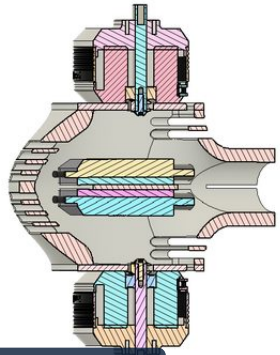
WHAT I DID

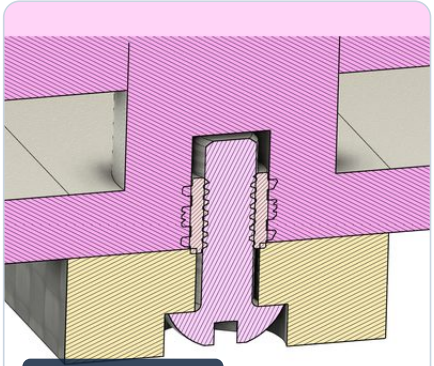
- Designed the 40 kg MTOW UAV with a 7 kg design payload and an 8+1 coaxial propulsion layout.
- Developed modular, 3D-printable airframe structures, motor mounts with ESC housings, landing skids, and avionics, battery, and payload bays.
- Applied DfAM and assembly design principles.
- Produced dimensioned three-view drawings and detailed CAD for fabrication and assembly.

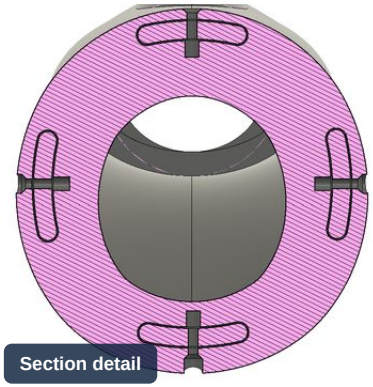



Exploded assembly


Transparent airframe layout


Motor nacelle section


Joint section detail


Section detail

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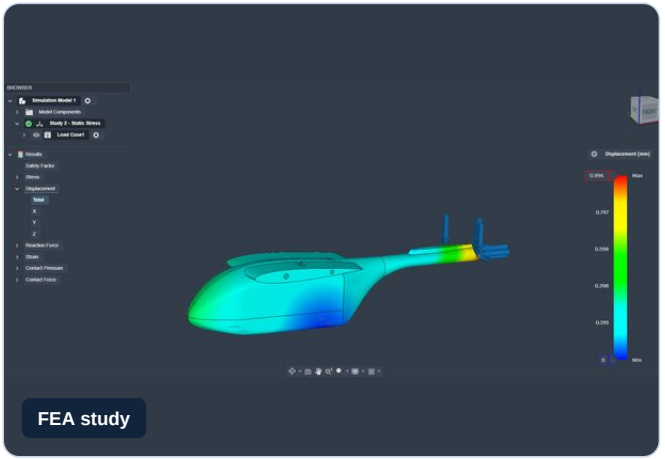
Detail design and structural checks

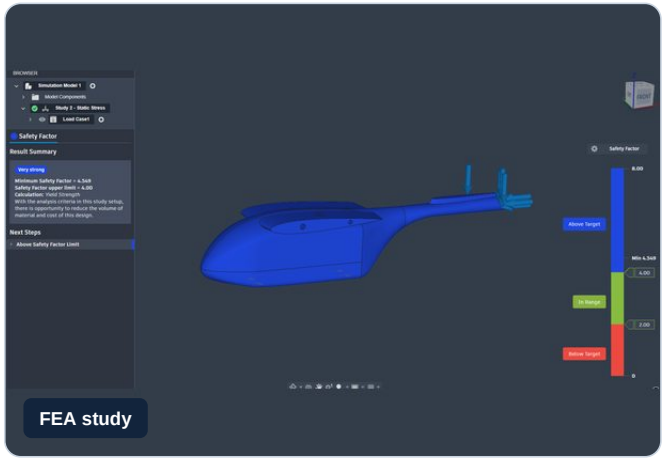
Modular parts and FEA studies that guided the printable airframe.

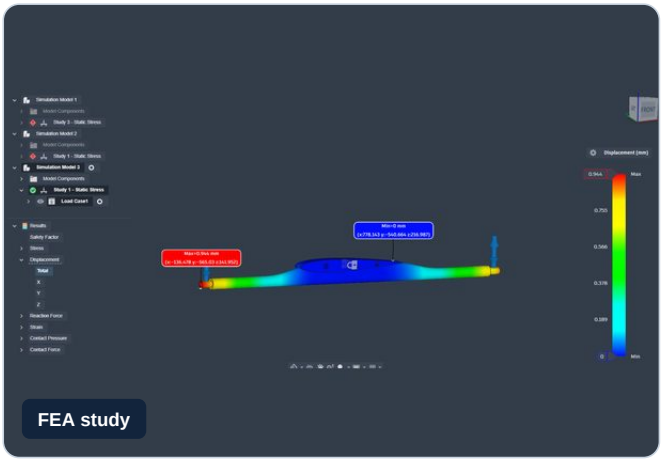
TOOLS Fusion 360 FEA studies

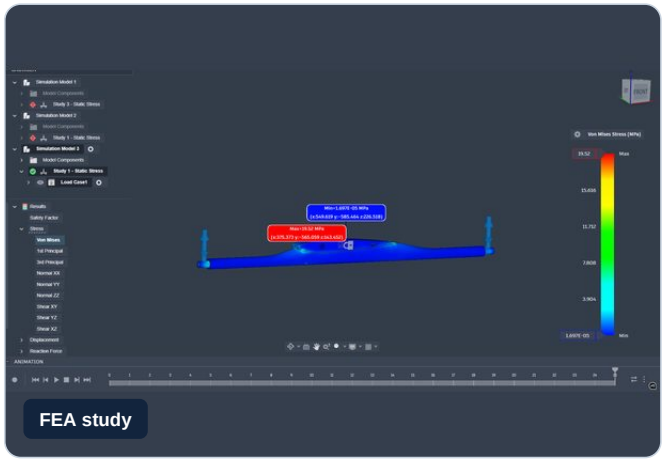
HIGHLIGHTS

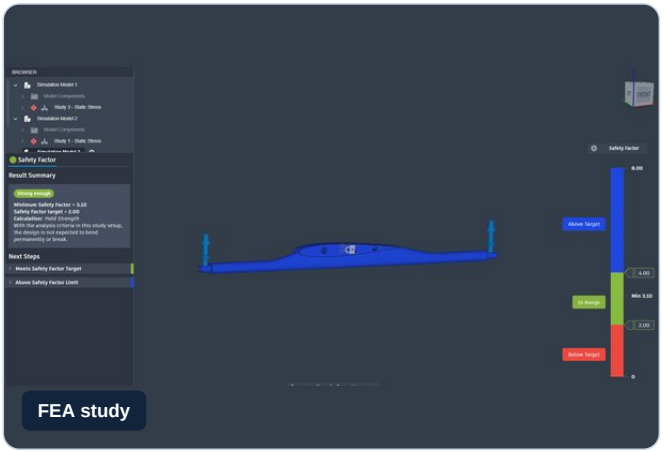
- **DfAM:** airframe split into modular, printable parts with assembly in mind.
- **Basic FEA** on motor arms, payload mounts, and tail components.
- **Motor hardware:** mounts with integrated ESC housings and coaxial motor nacelles.


FEA study


FEA study


FEA study


FEA study


FEA study


Motor mount detail

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04 | MALLARD ENTERPRISES | 2023 - 2025

ME-1A Amphibious Aircraft

THE PROJECT

Conceptual CAD models and design configurations for a 19-passenger amphibious aircraft, built in Fusion 360 and Blender with OpenVSP for configuration studies.

MISSION VARIANTS

Passenger, cargo, and military special-operations configurations developed from the same baseline aircraft.

MY ROLE

Aircraft Design Engineer and Assistant Project Manager (Jun 2024 - Sep 2025). Earlier, as design intern, led a four-person team on early conceptual design. Reviewed design reports and trade-offs, and coordinated vendors and RFP/RFQ responses.

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05 | NACDeC TEAM SKYPIAN | 2022 - 2023

Autonomous Air Ambulance eVTOL

Team concept for an air ambulance eVTOL with a cabin laid out around the patient.

Team of 5
led as team lead

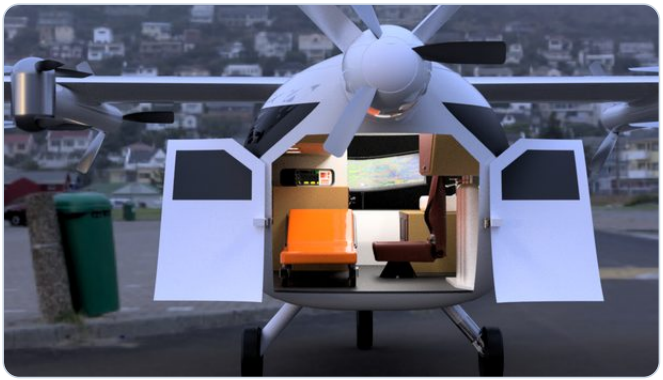
NACDeC
design competition

ROLE Team Lead and Aircraft CAD Designer (Oct 2022 - Jul 2023)
TOOLS Fusion 360, OpenVSP

WHAT I DID

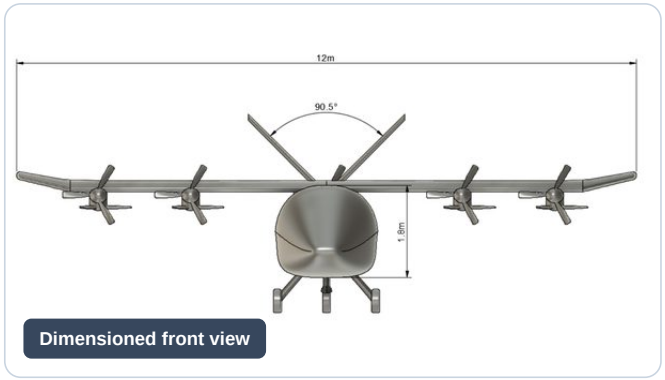
- Led a five-person team and modeled the aircraft structure and cabin layout.
- Optimized the cabin for patient transfer, medical equipment integration, and crew safety.
- Performed weight and CG estimates, aerodynamic studies, and integration checks.
- Prepared technical presentations for reviews, including a presentation at the Advanced Air Mobility Institute.











Dimensioned front view

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06 | CONCEPT STUDY

Supersonic UCAV

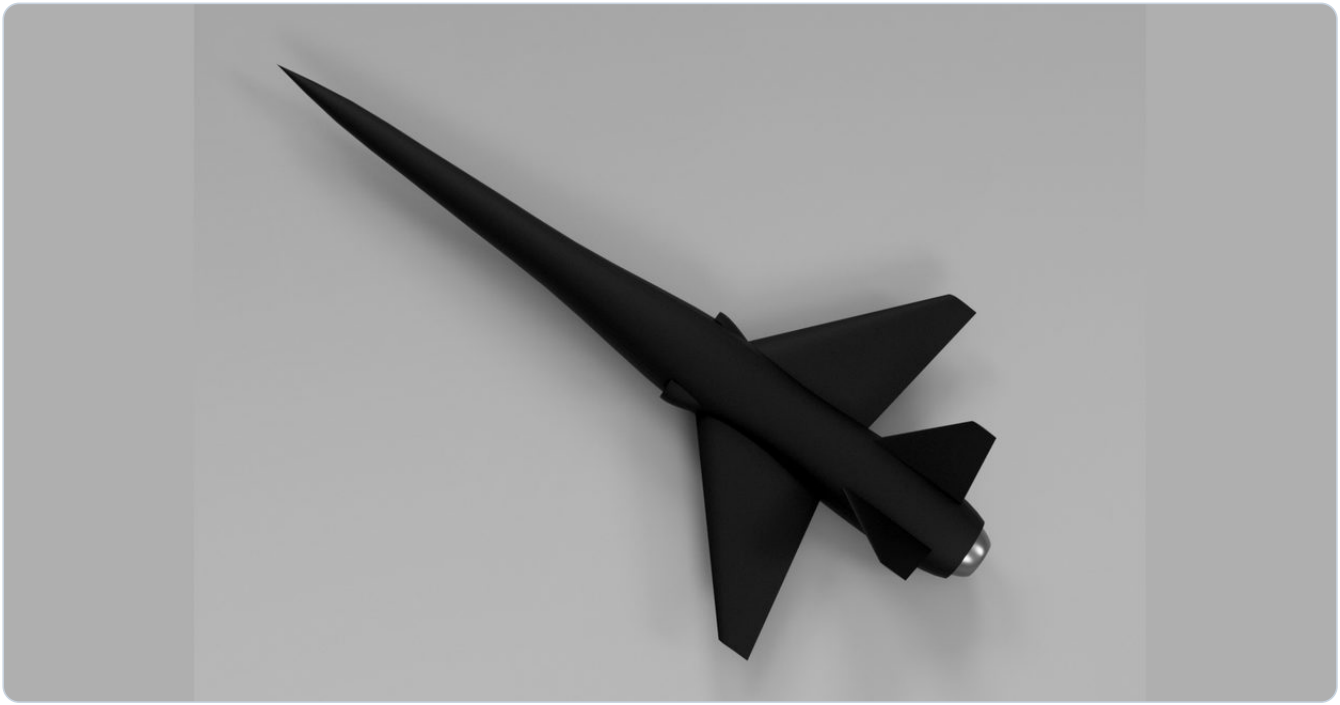
Conceptual design of a supersonic unmanned combat aerial vehicle for long-range strike.

Supersonic
long-range strike

ROLE Concept Designer
TOOLS Fusion 360, OpenVSP, mission analysis tools

WHAT I DID

- Developed the conceptual design for long-range strike missions, survivability, and high-speed performance.
- Focused the study on aerodynamic efficiency, mission requirements, and system-level integration.





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07 | CONCEPT STUDY

Passenger eVTOL Concept

A 3D concept for urban and regional air mobility, studied in several configurations.

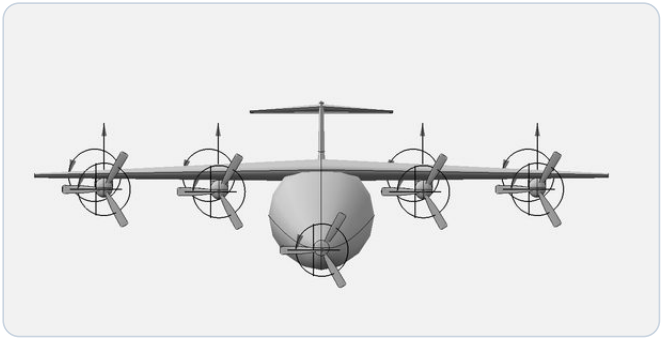
eVTOL
urban and regional

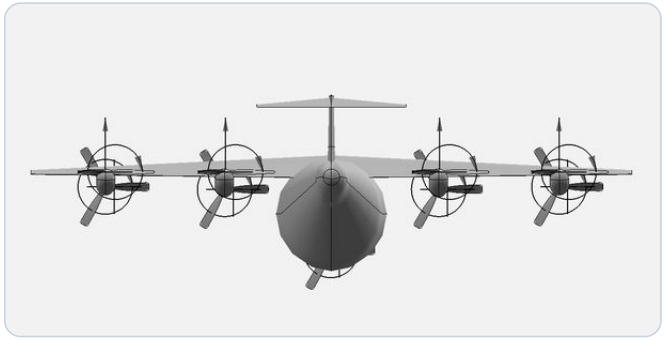
ROLE Lead Concept Designer
TOOLS Fusion 360, OpenVSP

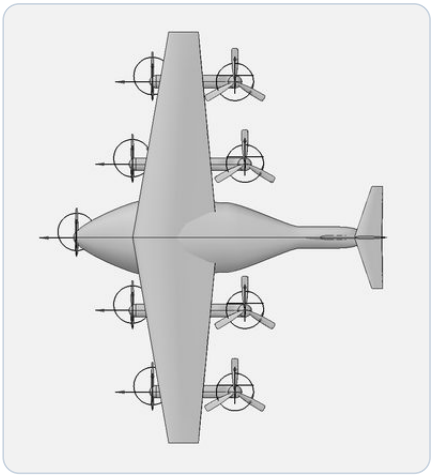
WHAT I DID

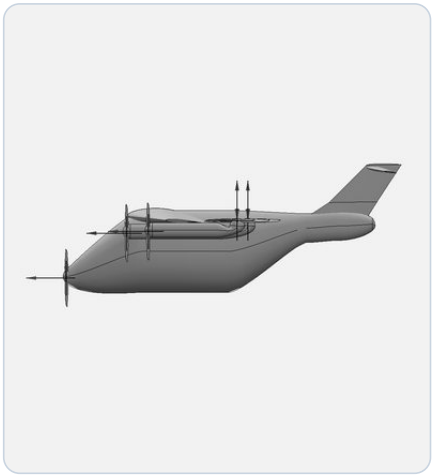
- Led the concept design of a passenger eVTOL aircraft for urban and regional air mobility.
- Built the 3D concept model and documented it in multiple views.









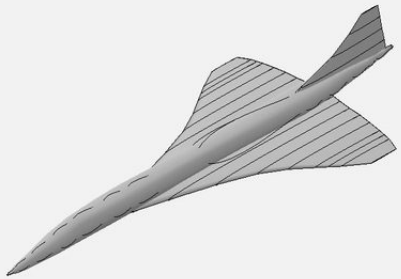




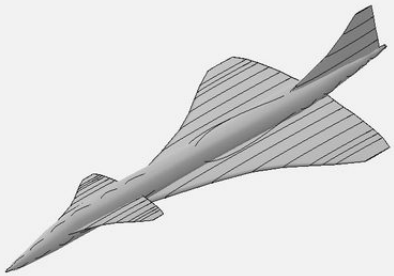
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Modified Concorde Aircraft

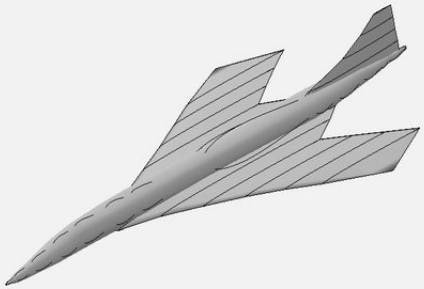
Role: Freelance Designer | Tool: OpenVSP. I developed multiple conceptual configurations of a modified Concorde for an international client and ran aerodynamic analyses on each, delivering results that combined technical depth with design clarity. Pressure-coefficient plots below are at Mach 2.04.



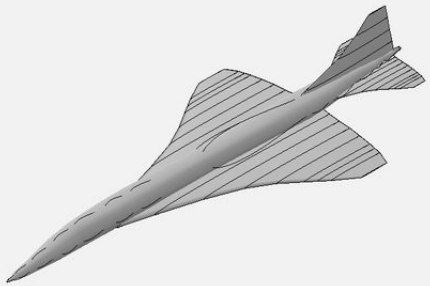
Configuration A



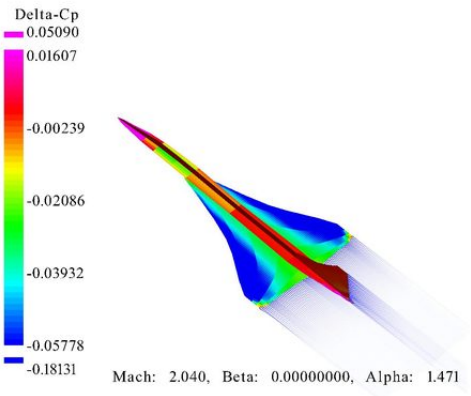
Configuration B



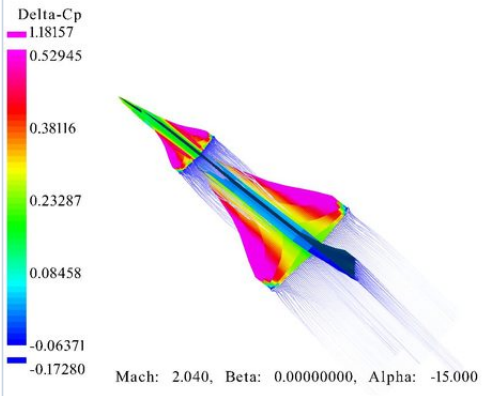
Configuration C



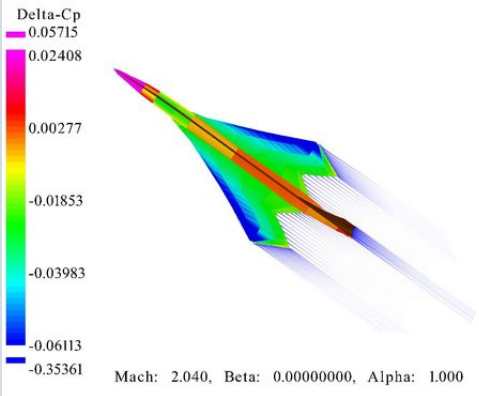
Configuration D



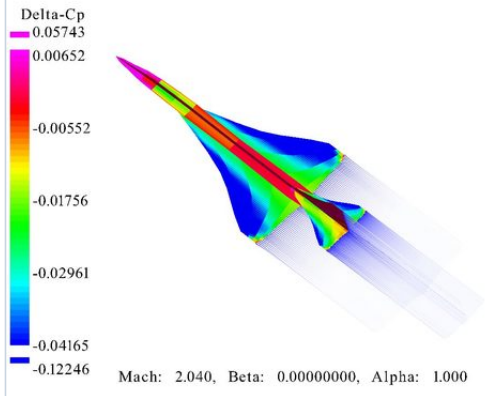
ΔCp, α = 1.5°



ΔCp, α = -15°



ΔCp, α = 1.0°



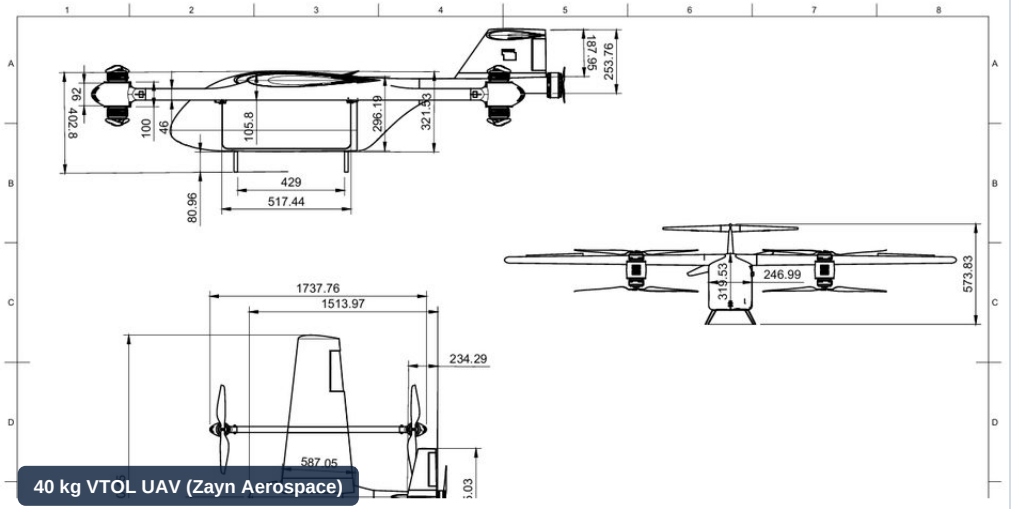
ΔCp, α = 1.0°

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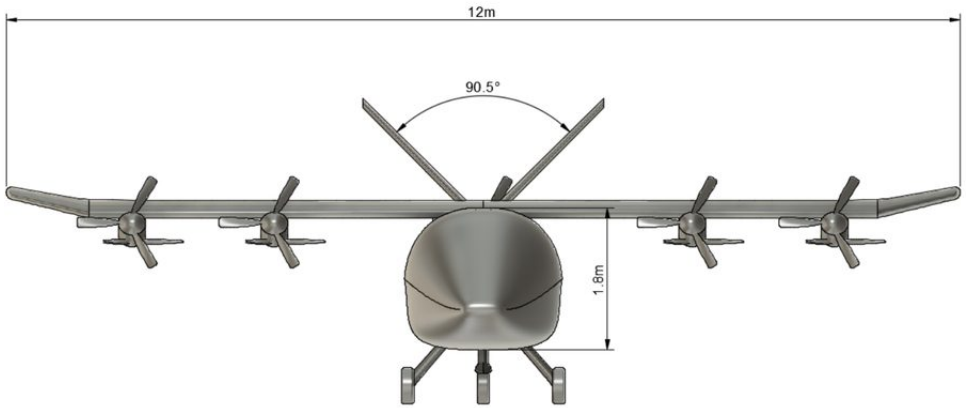
ENGINEERING DRAWING SAMPLES

Dimensioned three-view drawings

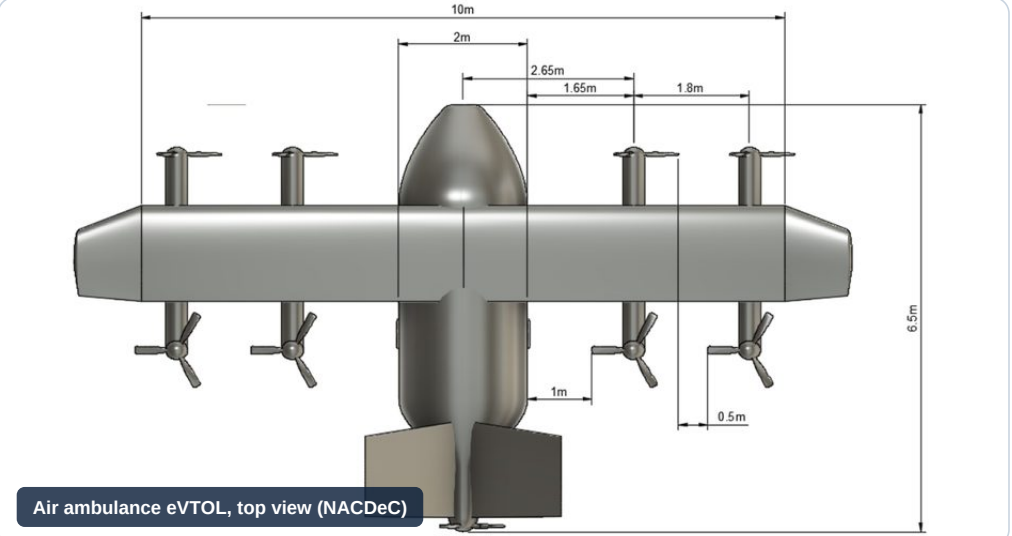
Overall dimensions and layouts prepared from the CAD models for fabrication, assembly, and design-review communication.



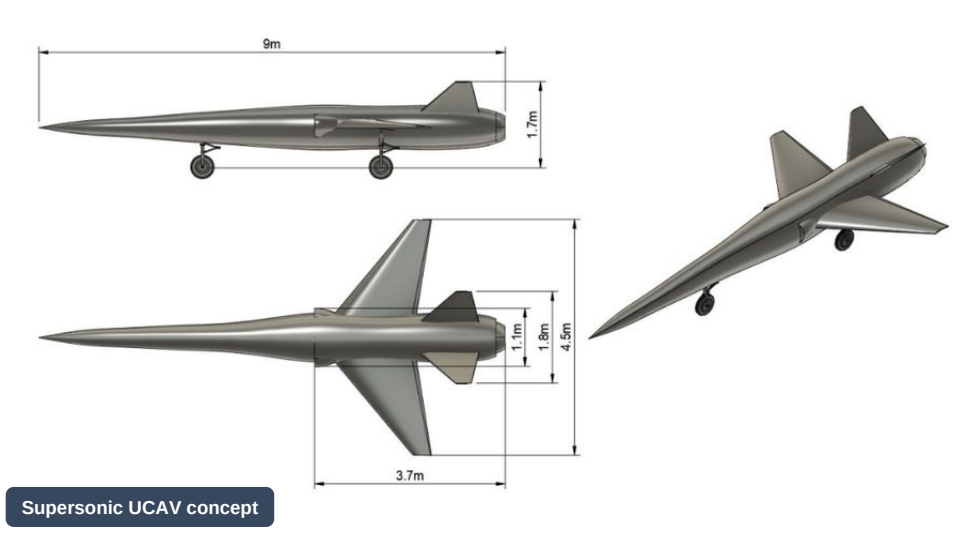
40 kg VTOL UAV (Zayn Aerospace)



Air ambulance eVTOL, front view (NACDeC)

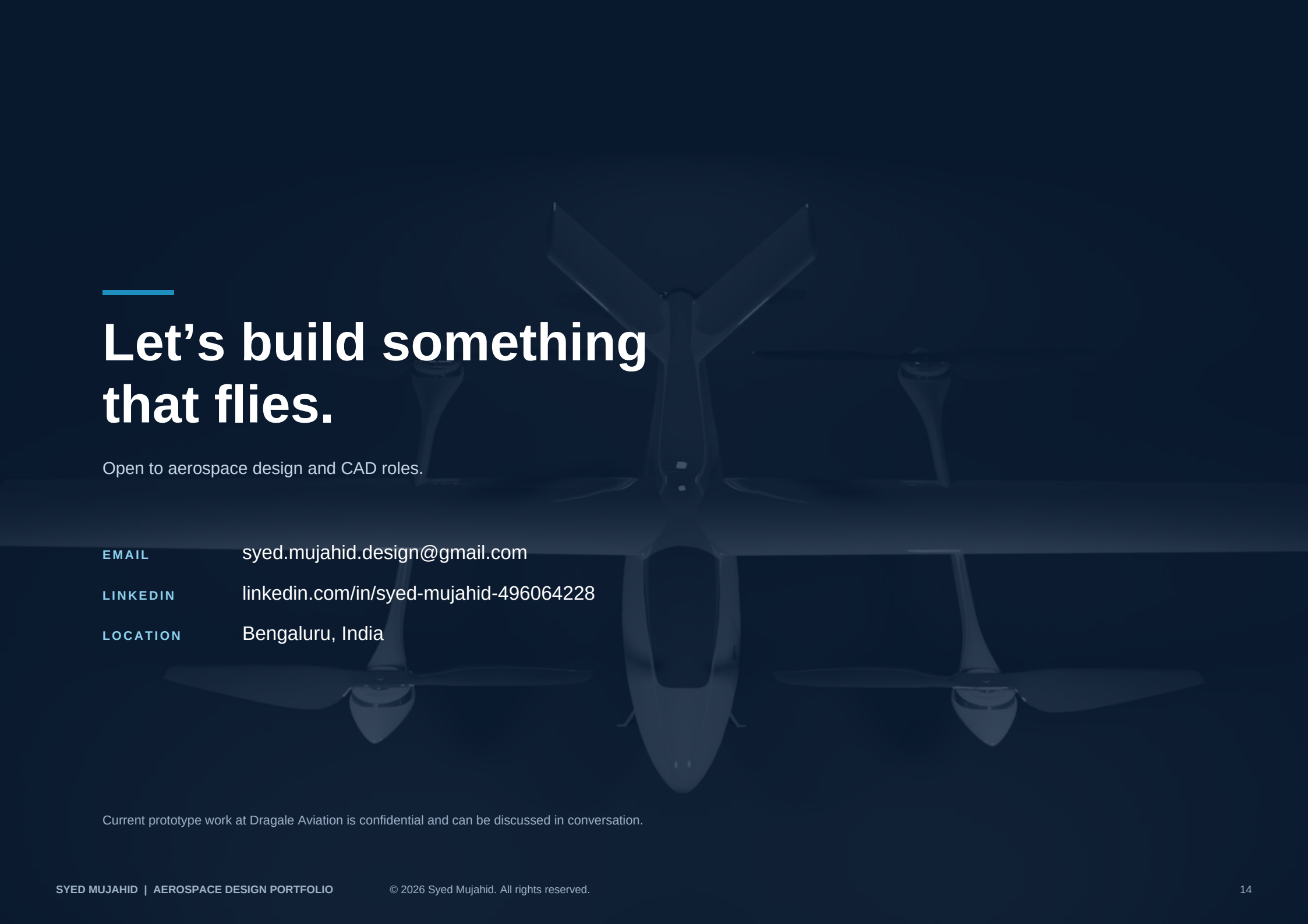


Air ambulance eVTOL, top view (NACDeC)



Supersonic UCAV concept

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Let’s build something that flies.

Open to aerospace design and CAD roles.

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LINKEDIN linkedin.com/in/syed-mujahid-496064228

LOCATION Bengaluru, India

Current prototype work at Dragale Aviation is confidential and can be discussed in conversation.

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